

Introduction to neural networks

Training aspects

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Contents

- 1 Performance measurement
- 2 Dataset split
- 3 Performance metrics
- 4 Overfitting
- 5 Regularization

Roadmap

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- 2 Dataset split
- 3 Performance metrics
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Performance measurement

- The central challenge in machine learning is that we must perform well on new, **previously unseen** inputs.



- The ability to perform well on previously unobserved inputs is called **generalization**.
 - Dataset split
 - Metric
 - Significance

¹ <https://www.bbc.com/news/world-us-canada-43604440>

² <https://www.formacar.com/en/tesla-63631-article>

Performance measurement

We will introduce a validation step and a test in order to measure generalization.

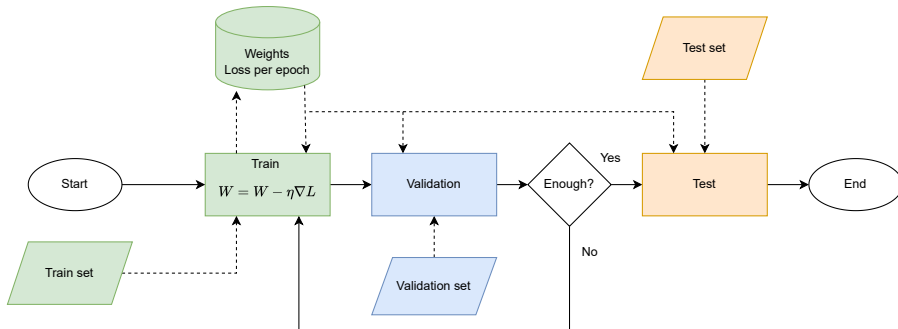


Figure: Train and test workflow

Roadmap

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Dataset split

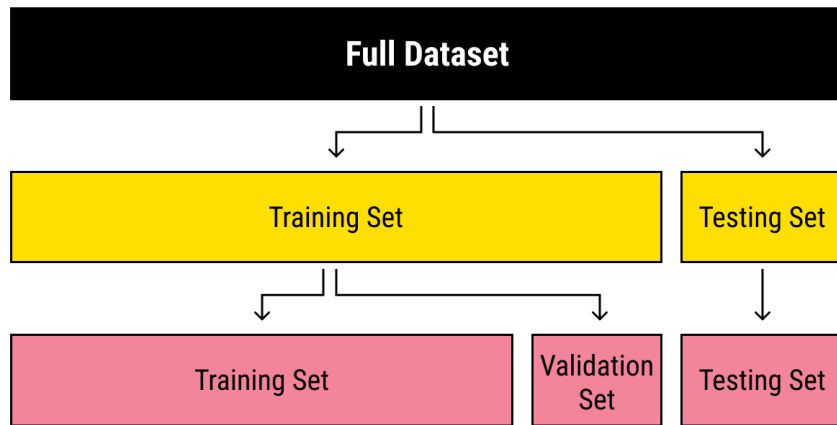


Figure: Dataset split [HJC]

Training and validation set split

N-fold validation

- Lets suppose a dataset, $X_{\Omega} = \{x_1, \dots, x_m\}$, with positive examples
- We could use a subset of our set of observations as an estimate of the performance

$$X = x_1, \dots, x_n$$

$$X' = x_{n+1}, \dots, x_m$$

, X is the **training set** and X' is usually called the **validation set** [Smola].

- Typically, one uses about 80% of the training data for training and 20% for validation [GoodFellow].

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Common measurements for categorical outputs

- Accuracy
- Precision
- Recall
- F1 score

Confusion matrix

- Computation:

Ground truth	Positive Prediction	Negative Prediction
Positive Class	TP	FN
Negative Class	FP	TN

Table: Confusion matrix

- $TP + FN$ sums 100 % of the positive class
- $FP + TN$ sums 100 % of the negative class

Example

Animal	 Barn owl  Chihuahua  Sheep dog
Not animal	 Apple  Muffin  Mop

Figure: Dataset of animals and no animals. ³

³<https://www.labelf.ai/blog/what-is-accuracy-precision-recall-and-f1-score>

Prediction concepts

For binary classes⁴

Network input	Prediction	Concept
	'animal'	true positive
	'not animal'	false negative
	'animal'	false positive
	'not animal'	true negative

⁴Consider the class 'animal' as the positive class.

Confusion matrix

		Predicted			
		Animal	Not animal		
Actual	Animal			True Positives	
	Not animal			True Negatives	
				False Positives	
				False Negatives	

Figure: Confusion matrix example.

Confusion matrix

		Predicted			
		Animal	Not animal		
Actual	Animal			True Positives	2
	Not animal			True Negatives	2
				False Positives	1
				False Negatives	1

Figure: Confusion matrix example.

Performance metrics

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

TP represents true positives, TN true negatives, FP false positives, and FN false negatives.

Performance metrics

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

In this formula, TP represents true positives and FP false positives. This metric measures the proportion of positive identifications that were actually correct.

Performance metrics

Recall

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

TP represents true positives and FN false negatives. Recall measures the model's ability to find all relevant positive instances in the dataset.

Performance metrics

F1-score

The F1-score, which is the harmonic mean of precision and recall, can be expressed as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Thus, the F1-score combines both precision and recall into a single metric that accounts for both false positives and false negatives. It is especially useful in situations where the balance between precision and recall is critical.

Significance





Definition

A result is statistically significant if the probability of it occurring by chance is very low, typically defined by a threshold α (e.g., 0.05).

Core Concepts:

- **Null Hypothesis (H_0):** No real difference between models.
- **p-value:** $P(\text{Data}|H_0)$. If $p < \alpha$, we reject H_0 .

Application in ML (Model Comparison): To compare error rates ϵ_1 and ϵ_2 using a Z-test:

$$Z = \frac{|\hat{\epsilon}_1 - \hat{\epsilon}_2|}{\sqrt{\frac{\hat{\epsilon}(1-\hat{\epsilon})}{n_1} + \frac{\hat{\epsilon}(1-\hat{\epsilon})}{n_2}}} \quad (5)$$

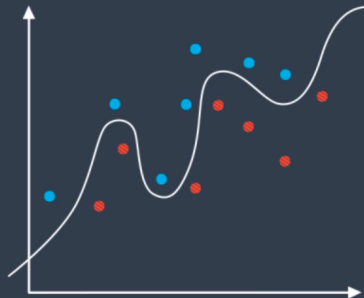
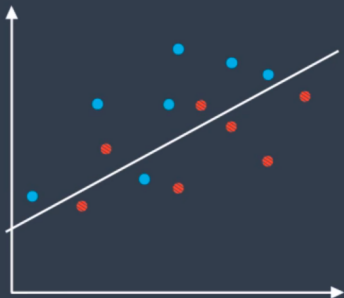
Ensures that performance gains are robust and reproducible.

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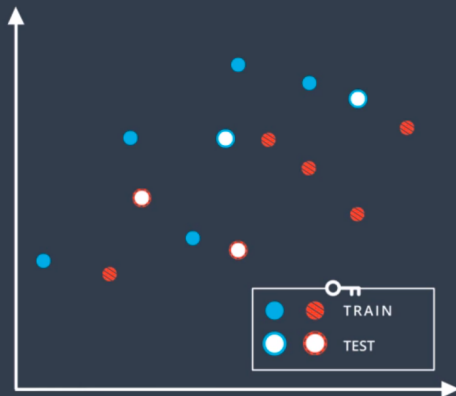
Overfitting problem

Which model is better?



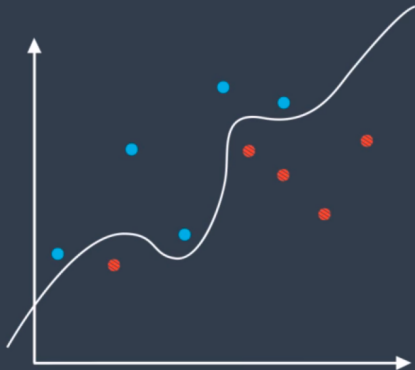
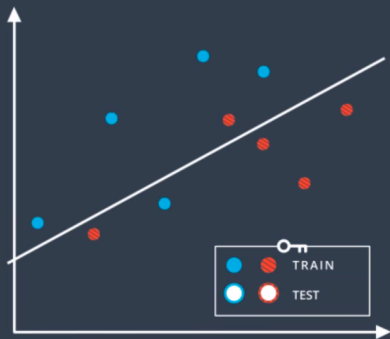
Overfitting problem

The importance of testing



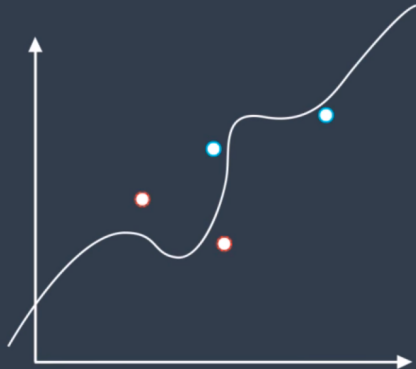
Overfitting problem

The importance of testing



Overfitting problem

The importance of testing



Overfitting and generalization

- **Underfitting** occurs when the model is not able to obtain a sufficiently low error value on the training set.
- **Overfitting** occurs when the gap between the training error and test error is too large.
- A model's **capacity** is its ability to fit a wide variety of functions.

Regularization

Identify overfitting

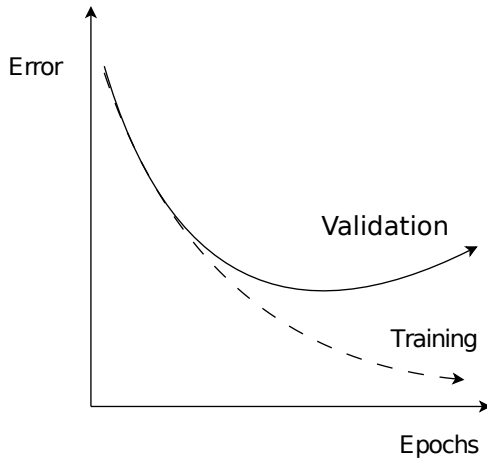


Figure: Loss as epochs advance.

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- **Regularization** is any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.
 - Early termination
 - L1, L2
 - Dropout

Regularization

Early termination

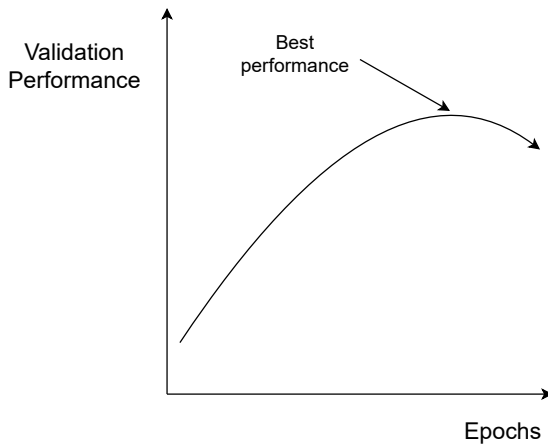


Figure: Performance as epochs advance.

L2 Regularization (Weight Decay)

Adds a penalty equal to the square of the magnitude of coefficients.

The Math:

$$\mathcal{L}(W) = E(W) + \lambda \sum_{j=1}^n w_j^2$$

Key Insights:

- Penalizes large weights.
- Leads to **small, non-zero** weights.

L1 Regularization (Lasso)

Adds a penalty equal to the absolute value of the magnitude of coefficients.

The Math:

$$\mathcal{L}(W) = E(W) + \lambda \sum_{j=1}^n |w_j|$$

Key Insights:

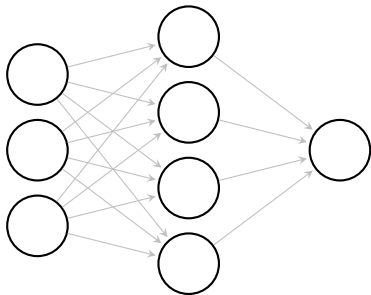
- Encourages **sparsity** (many weights become exactly zero).
- Useful for **feature selection**.

A stochastic technique mainly used in Deep Learning.

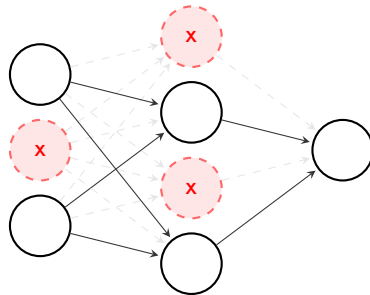
- **How it works:** During training, randomly "deactivate" a percentage p of neurons in each forward pass ⁵.
- **Why it works:**
 - Prevents **co-adaptation** of neurons.
 - Forces the network to learn redundant representations.
 - Acts as an approximate model ensemble (averaging many sub-networks).

⁵All neurons are used during inference, but weights are scaled by p .

Standard Network



Network with Dropout



Data Augmentation

- Artificially increases the size and diversity of the training set by generating new examples from the existing ones.
- Acts as a regularizer: it reduces overfitting and improves generalization, especially with small datasets.
- New samples are created by applying label-preserving transformations:

$$\mathbf{x}' = A(\mathbf{x}; \phi), \quad y' = y \quad (6)$$

$$\phi \sim p(\phi)$$

Parameters ϕ (e.g., rotation angle, noise level) are sampled randomly at training time. Unlike preprocessing, augmentation is applied **only to the training set**, never to validation/test data.

Data Augmentation

Common techniques

- **Geometric transforms:** rotation, translation, scaling, horizontal/vertical flips, cropping.
- **Photometric transforms:** brightness, contrast, saturation changes, color jittering.
- **Noise injection:** adding Gaussian noise,

$$\mathbf{x}' = \mathbf{x} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (7)$$

- **Erasing/occlusion:** randomly masking regions of the input (random erasing, cutout).
- Transformations must preserve the label: e.g., a vertical flip may be invalid for letters ('A' vs 'V') or digits ('6' vs '9').

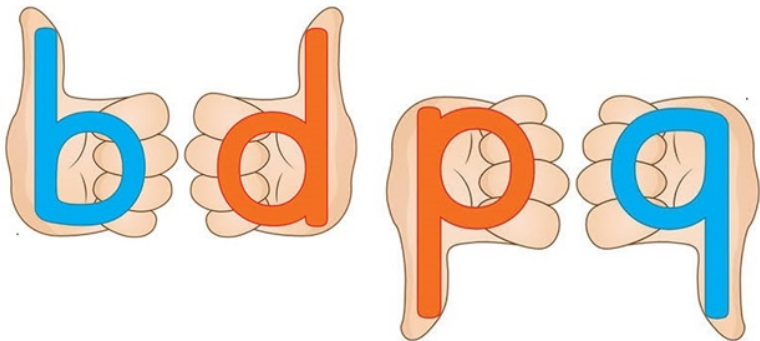


Figure: Rotations and transforms

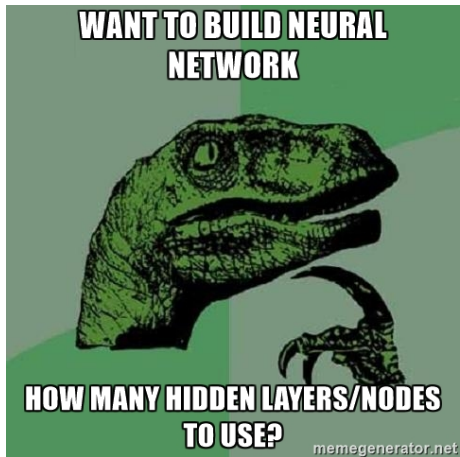
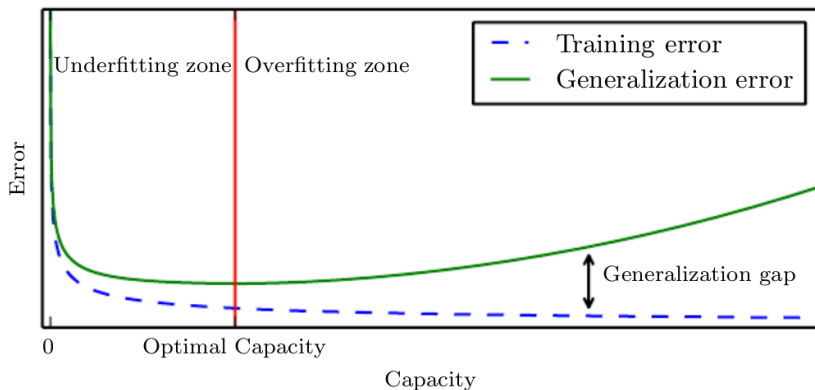


Figure: Selecting depth

Model capacity selection

- Usually select the simplest model
- The optimal model is the one that has the same capacity than the problem.



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